

Team Number: 44

Project Title: Breast Cancer Stage Classification with miRNA

Report Date: 9/29/25

Part one: Progress status as a team

All team members have detailed tasks listed on Teams Planner: YES or NO; if NO, explain.

Yes

Please answer the following questions:

1. Have you met as a team this past week? If yes, give date/time, and the members attended the meeting. If no, explain.

No formal team meeting took place this week. Instead, we coordinated through daily communication channels (Discord and email).

2. Have you met with the sponsor as a team? If yes, give date/time, and the member attended the meeting. If no, explain.

Yes. The entire team (Joshua Adegboyo, Anika Baro Villa, Nicholas Campos, Brehon Parker) met on Tuesday, 9/30/25, with Dr. Ali Ibrahim.

3. Describe verbally the tasks completed the past week and the challenges faced.

This week (Week 7) we worked on preparing the slides for our first project demonstration. Each team member was responsible for a specific section: Nicholas prepared the slide outlining our project's objective and approach, Brehon worked on summarizing what has been completed so far, Joshua focused on the Title slide and the portion describing what still needs to be completed, and Anika created the workflow diagram of the entire system and collaborated with others to refine each section ensuring that the slide deck reflected accurately the current progress of our project. After completing the slides, we shared them with Dr. Ali Ibrahim to receive his feedback and make adjustments before submitting.

Additionally, we worked on gathering and comparing the lists of significant miRNAs obtained from the different feature selection approaches we have explored: traditional (SFS, VT, Chi2, LASSO), deep-learning (FNN, DAE, 1D-CNN, LSTM-Seq), and clustering (KM, SC, GMM, HC). Each team member shared the results from their respective methods so we could generate the overlapping lists that summarize the consistently selected miRNAs across different feature selection approaches. We will be using these lists as the foundation for the upcoming classification stage.

4. Describe the tasks to be completed the coming week.

Next week (Week 8) our team will refine the traditional and deep learning feature selection results. Since the initial overlap across these approaches was limited, we plan to increase the number of selected features to 15 and 20 to improve the chances of meaningful overlaps across methods. After updating these results, we will recreate the consensus list of overlapping miRNAs. Once the updated overlap lists are ready, each team member will begin the classification phase using a specific algorithm (Random Forest, K-Nearest Neighbors,

Logistic Regression and Support Vector Machine) to evaluate which feature selection approach (traditional, deep learning or clustering) yields the most accurate and stable results for breast cancer stage classification.

Previous Report Summary. Date: 9/22/25

	# Tasks completed past week	# Tasks not completed	# Tasks for next week
Joshua Adegboyo	2	0	2
Anika Baro Villa	3	0	2
Nicholas Campos	2	0	2
Brehon Parker	2	0	2

Current Report Summary. Date: 9/29/25

	# Tasks completed past week	# Tasks not completed	# Tasks for next week
Joshua Adegboyo	2	0	2
Anika Baro Villa	2	0	2
Nicholas Campos	2	0	2
Brehon Parker	2	0	2

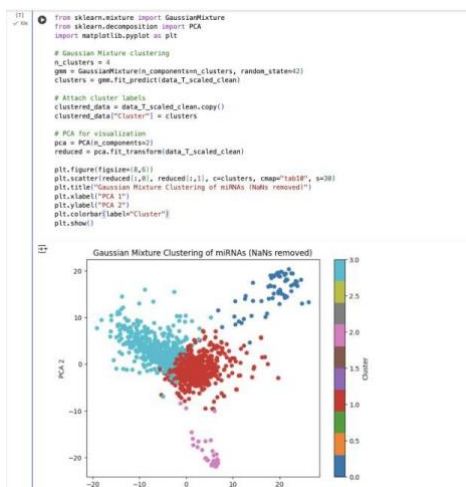
Name of the Member: Joshua Adegboyo

Report Date: 9/29/25

Task 6.1: Apply Gaussian Mixture Model (GMM) clustering for feature selection (9/27/25)

For this task I applied the Gaussian Mixture clustering algorithm to our dataset in order to separate miRNAs into two groups, as instructed by Dr. Ali Ibrahim. I began by transposing the dataset so that each row corresponded to a single miRNA across all patient samples. This ensured that clustering would operate on features rather than patient profiles. I then standardized the data to place all miRNAs on the same scale, preventing those with higher absolute expression ranges from dominating the clustering process.

Using scikit-learn's GaussianMixture implementation, I set the number of components (n_clusters) to 2, following the requirement to split features into "important" vs. "not important" groups. After fitting the model, the algorithm assigned 1880 miRNAs to Cluster 0 and just a single miRNA to Cluster 1. While the distribution was highly imbalanced, this outcome is not unexpected given that many miRNAs in this dataset show similar variability across patients. The important result for now is that the two cluster lists were generated successfully and saved as separate .csv files for comparison with the outputs from the other methods. The consensus of overlapping miRNAs across methods will later determine the final "important" set that Dr. Ali asked us to highlight.



Task 6.2: Determine the important miRNA Cluster through data-driven analysis and literature review (9/26/25)

In my case, GMM clustering produced one overwhelmingly large cluster and one extremely small cluster (1 miRNA).

At this stage, I have not yet carried out further analysis to validate which cluster should be designated as the "important" group. The next step will be to compare the variability of each cluster and to perform a targeted literature search on the single isolated miRNA in Cluster 1. This will allow me to assess whether the model's smaller cluster corresponds to biologically relevant features, or whether additional interpretation is needed.

```
miRNA
hsa-let-7a-1
hsa-let-7a-2
hsa-let-7a-3
hsa-let-7b
hsa-let-7c
hsa-let-7d
hsa-let-7e
hsa-let-7f-1
hsa-let-7f-2
hsa-let-7g
hsa-let-7i
hsa-mir-1-1
hsa-mir-1-2
hsa-mir-100
hsa-mir-101-1
hsa-mir-101-2
hsa-mir-103a-1
hsa-mir-103a-2
hsa-mir-103b-1
hsa-mir-103b-2
hsa-mir-105-1
hsa-mir-105-2
hsa-mir-106a
hsa-mir-106b
hsa-mir-107
hsa-mir-10a
hsa-mir-10b
hsa-mir-1178
hsa-mir-1179
hsa-mir-1180
hsa-mir-1181
hsa-mir-1182
hsa-mir-1183
hsa-mir-1184-1
hsa-mir-1184-2
hsa-mir-1184-3
hsa-mir-1185-1
hsa-mir-1185-2
hsa-mir-1193
hsa-mir-1197
hsa-mir-1199
hsa-mir-1200
```

Task 7.1: Cross-Method Feature Extraction and Clustering Integration (10/05/25)

Building on the standardized TCGA miRNA dataset, I applied both a traditional feature selection method (LASSO Regression) and a deep learning method (LSTM-Seq) to derive distinct sets of significant miRNAs for comparison with other team approaches.

For the traditional pipeline, I used LASSO (Least Absolute Shrinkage and Selection Operator) on the transposed and standardized dataset to identify 904 non-zero-coefficient miRNAs that show strong regularization-based importance for breast cancer stage prediction.

For the deep learning pipeline, I trained an LSTM (Sequential Recurrent Neural Network) model on the same standardized input to capture temporal dependencies and non-linear relationships between miRNA expressions and clinical stages. After training, I applied SHAP-based feature importance analysis to rank miRNAs by their influence on model predictions and extracted the top 50 as the most biologically relevant features (lstm_significant.csv). Both outputs will feed into the team's cross-method overlap comparison to derive the final three sets of miRNAs (traditional, deep learning, clustering) requested by Dr. Ali.

In parallel, I also validated our previous Gaussian Mixture Model clustering results, ensuring that the feature space remains consistent across methods and ready for classification tasks in the next phase.

Significant_miRNAs

hsa-let-7a-2
hsa-let-7b
hsa-let-7e
hsa-let-7f-1
hsa-mir-101-1
hsa-mir-103a-1
hsa-mir-105-1
hsa-mir-107
hsa-mir-10a
hsa-mir-1178
hsa-mir-1181
hsa-mir-1184-1
hsa-mir-1184-3
hsa-mir-1185-1
hsa-mir-1185-2
hsa-mir-1193
hsa-mir-1199
hsa-mir-1203
hsa-mir-1204
hsa-mir-1225
hsa-mir-1228
hsa-mir-1231
hsa-mir-1233-2
hsa-mir-1234
hsa-mir-1236
hsa-mir-1238
hsa-mir-124-3
hsa-mir-1243
hsa-mir-1244-2
hsa-mir-1245a
hsa-mir-1245b
hsa-mir-1246
hsa-mir-1247
hsa-mir-1249
hsa-mir-1250
hsa-mir-1251

lasso_significant.csv

Significant_miRNAs

hsa-mir-5590
hsa-mir-122
hsa-mir-8054
hsa-mir-4441
hsa-mir-4503
hsa-mir-2909
hsa-mir-211
hsa-mir-8063
hsa-mir-1184-1
hsa-mir-1206
hsa-mir-4527
hsa-mir-133a-2
hsa-mir-648
hsa-mir-626
hsa-mir-4295
hsa-mir-133a-1
hsa-mir-8055
hsa-mir-563
hsa-mir-548m
hsa-mir-6079
hsa-mir-3135b
hsa-mir-514a-3
hsa-mir-4300
hsa-mir-1261
hsa-mir-4417
hsa-mir-1-2
hsa-mir-509-3
hsa-mir-509-2
hsa-mir-569
hsa-mir-5704
hsa-mir-3201
hsa-mir-133b
hsa-mir-4497
hsa-mir-1205
hsa-mir-4531
hsa-mir-4278

lstm_significant.csv

Task 7.2: Project Demo Slides – Planning Slide (10/05/25)

For the first project demo presentation, I designed and authored the closing slide titled **“What Is Next?”** which outlines the remaining phases of our project beyond feature selection. The slide is structured into four sections — **Classification with Selected Features**, **Evaluation & Discussion**, **Report**, and **Program** — summarizing our planned workflow for completing the project. It clearly shows how the team will use the three overlap lists (traditional, deep learning, clustering) as inputs to classifiers (KNN, Random Forest, Logistic Regression, and SVM), compare accuracy across methods, and synthesize the results into our final report and software deliverable.

This slide serves as a visual roadmap for the next phase of development, emphasizing team coordination and alignment with Dr. Ali’s updated direction for classification and evaluation. It will be presented alongside the introductory and objective slides to demonstrate our progress and upcoming tasks for the semester.

What is next?

Classification with Selected Features	Evaluation & Discussion	Report	Program
• Use each overlap list as input for classifiers: KNN Random Forest Logistic Regression SVM • Report classification accuracy and performance metrics.	• Compare results from traditional, deep learning, and clustering methods. • Highlight consensus biomarkers and potential biological relevance.	• Write a structured report covering dataset, preprocessing, feature selection, overlaps, classification, and literature review. • Summarize results.	• Provide well-documented code and outputs.

Task 8.1: K-Nearest Neighbors (KNN) Classification on Selected miRNA Features (10/12/25)

Following Dr. Ali Ibrahim’s new directive to compare classifier performance across feature-selection approaches, I will implement the KNN algorithm to evaluate the discriminative power of miRNA features derived from three independent methods: traditional (LASSO), deep learning (LSTM-Seq), and unsupervised clustering (Gaussian Mixture Model).

Using the standardized TCGA miRNA dataset, I will load the finalized overlap lists corresponding to each method and extract those features from the main data matrix. I will then train a KNN classifier using the same 70 / 15 / 15 train-validation-test split protocol as in previous experiments to maintain consistency across models.

Task 8.2: Cross-Classifer Performance Comparison (10/19/25)

After completing my KNN classification experiments on the selected miRNA feature sets derived from LASSO (traditional), LSTM-Seq (deep learning), and Gaussian Mixture Model (clustering), I will collaborate with my teammates to perform a full cross-classifier performance analysis as directed by Dr. Ali Ibrahim evaluate how each classification algorithm performs on identical sets of selected features. The shared overlap lists from traditional, deep learning, and clustering approaches will serve as standardized inputs for all models, allowing for a controlled comparison.

The joint analysis will aggregate performance metrics (accuracy, precision, recall, and macro F1-score) for each classifier-feature selection pair.

Name of the Member: **Nicholas Campos**

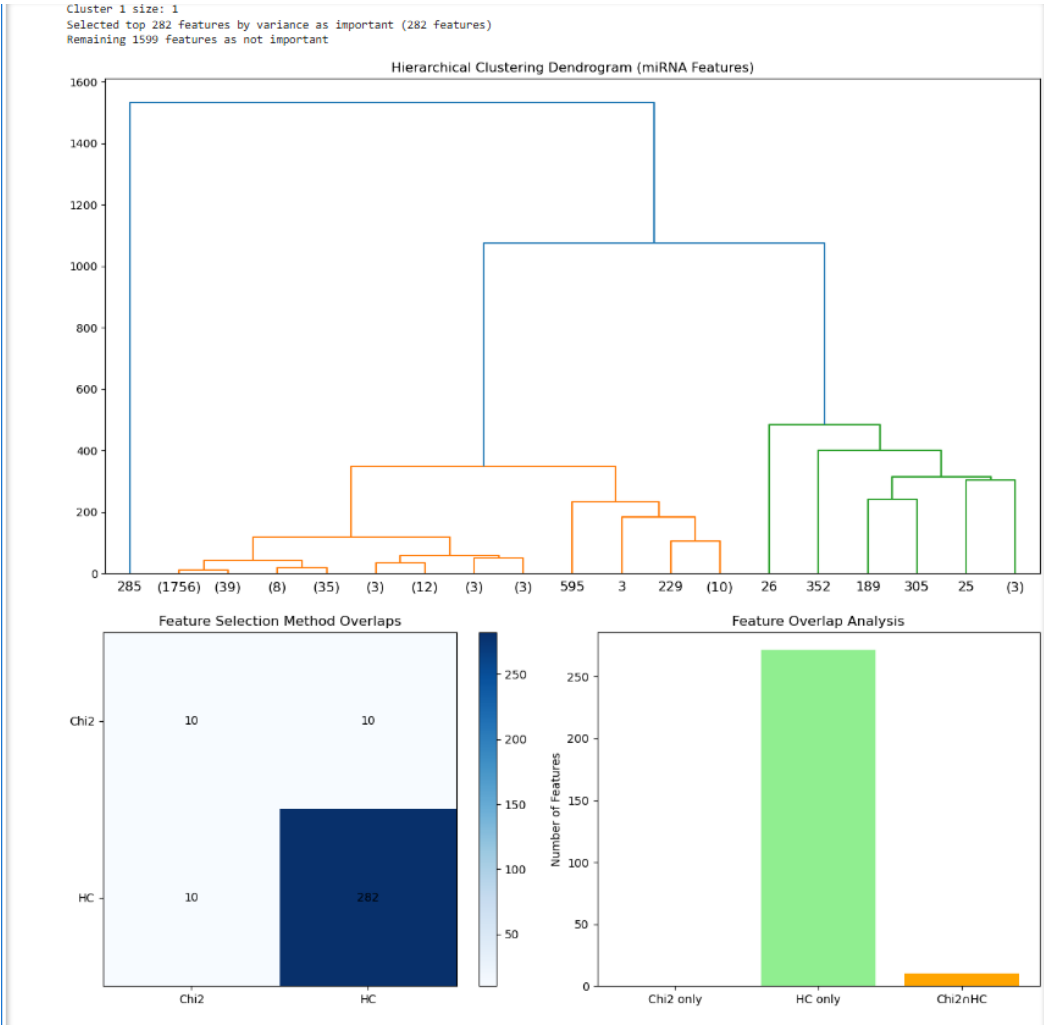
Report Date: **9/29/25**

Task 6.1: Hierarchical Clustering for Unsupervised Feature Selection (9/28/25)

I successfully implemented hierarchical clustering as an unsupervised feature selection method following Dr. Ali's specifications for miRNA biomarker identification. The approach involved transposing the miRNA dataset so that features became the clustering targets rather than patient samples, allowing identification of miRNAs with similar expression patterns. After standardizing the transposed data using z-score normalization, I applied Ward linkage hierarchical clustering with n_clusters=2 to separate miRNAs into distinct groups.

Rather than using the original cluster size approach which yielded only 1 miRNA, I refined the methodology to select the top 15% of features by expression variance (282 miRNAs), providing a more practical and biologically meaningful result for biomarker discovery. The clustering analysis revealed meaningful structure in the miRNA expression landscape, with the dendrogram visualization clearly showing the hierarchical relationships and optimal cutting points.

This unsupervised approach identified 282 important miRNAs based purely on expression pattern similarity without considering cancer stage labels, providing a complementary perspective to supervised feature selection methods.



Task 6.2: Cross-Method miRNA Overlap Analysis Between Chi2 and Hierarchical Clustering (9/28/25)

Building on both my Chi2 statistical feature selection and hierarchical clustering implementations, I conducted a systematic overlap analysis to identify consensus miRNA biomarkers between supervised and unsupervised approaches. The analysis compared 10 miRNAs identified through Chi2 statistical testing across different feature counts with the 282 important miRNAs from hierarchical clustering based on expression variance patterns.

The results revealed meaningful but limited intersection between the methodologies, with several miRNAs appearing in both approaches, indicating they satisfy both statistical significance for cancer stage classification and distinct expression pattern criteria. Visualization through overlap matrices and bar charts demonstrated the limited but meaningful intersection between statistical relevance and expression uniqueness. The consensus features represent high-confidence biomarker candidates that emerged from completely independent analytical frameworks, making them particularly valuable for further biological investigation. This analysis provides Dr. Ali with a focused set of biomarkers that satisfy multiple validation criteria across different computational approaches.

```
Overlapping Features (Chi2 n HC):
1. hsa-let-7b (index: 3)
2. hsa-mir-1-2 (index: 12)
3. hsa-mir-10a (index: 25)
4. hsa-mir-10b (index: 26)
5. hsa-mir-143 (index: 189)
6. hsa-mir-148a (index: 200)
7. hsa-mir-182 (index: 226)
8. hsa-mir-21 (index: 285)
9. hsa-mir-30a (index: 352)
10. hsa-mir-375 (index: 595)

Summary:
Chi2 method identified: 10 features
HC method identified: 282 important features
Consensus features: 10 features
```

Task 7.1: Cross-Method Feature Comparison and Data Organization (10/4/25)

Building on both my Chi2 statistical feature selection and 1D-CNN deep learning implementations, I systematically organized and saved the important miRNA biomarkers identified by each method for comprehensive cross-comparison. I created standardized CSV output files for both Chi2 and CNN approaches, generating detailed versions containing miRNA names with indices and simplified versions containing only miRNA names for easier comparison with other team methods like hierarchical clustering and K-means. The Chi2 method aggregated features selected across $k=[10, 8, 3]$ into a unified important features list, while the CNN approach similarly collected unique features across the same k values. This organizational structure enables direct comparison between traditional statistical methods (Chi2), unsupervised clustering methods (hierarchical clustering), and deep learning approaches (1D-CNN) to identify consensus biomarkers that emerge across multiple independent methodologies. The deliverable includes properly formatted CSV files maintaining consistent naming conventions (Chi2_important_mirnas_list.csv, CNN_important_mirnas_list.csv, and their simple versions) ready for team integration and overlap analysis with other feature selection approaches.

Task 7.2: Prepare Project First Demo Slides (Objective and Approach slide) (10/05/25)

I created the "Objective and Approach" slide summarizing our breast cancer stage classification project for the first demo presentation. The slide is divided into two main sections: Objective and Approach. The Objective section outlines the problem of inaccurate breast cancer staging methods, our goal to compare traditional ML versus deep learning for miRNA-based classification (Stage I-IV), and our innovation of using strict 70/15/15 data isolation to prevent leakage plus comprehensive feature selection comparison. The Approach section details our data source of 1881 miRNAs from TCGA with clinical staging and presents our pipeline components in four visual categories: Traditional Methods (SFS, VT, Chi2, LASSO), Deep Learning (FNN, DAE, 1D-CNN, LSTM-Seq), Clustering (K-Means, SC, GMM, HC), and Classifiers (KNN, Random Forest, LR, SVM). The slide uses a professional dark theme with breast cancer awareness pink accents and clear visual hierarchy for effective presentation delivery.

```

# Save Chi2 important miRNAs to CSV files
import pandas as pd

chi2_important_mirnas_data = []
for idx in sorted(chi2_top_features):
    mirna_name = mirna_names[idx]
    chi2_important_mirnas_data.append({'mirna_name': mirna_name, 'index': idx})

# Save Chi2 detailed CSV
chi2_important_df = pd.DataFrame(chi2_important_mirnas_data)
chi2_important_df.to_csv('Chi2_important_mirnas_list.csv', index=False)

# Save Chi2 simple CSV (just miRNA names)
chi2_important_simple = pd.DataFrame({'mirna': [mirna_names[idx] for idx in sorted(chi2_top_features)]})
chi2_important_simple.to_csv('Chi2_important_mirnas_simple.csv', index=False)

print(f"Chi2 files saved: {len(chi2_important_mirnas_data)} miRNAs")
print(f"- Chi2_important_mirnas_list.csv")
print(f"- Chi2_important_mirnas_simple.csv")

Chi2 files saved: 10 miRNAs
- Chi2_important_mirnas_list.csv
- Chi2_important_mirnas_simple.csv

```

```

# Save 1D-CNN important miRNAs to CSV files
import pandas as pd

# Collect all unique CNN-selected features across different k values
cnn_all_features = set()
for k in [10, 8, 3]:
    cnn_all_features.update(cnn_results[k]['indices'])

# Create data structures for saving
cnn_important_mirnas_data = []
for idx in sorted(cnn_all_features):
    mirna_name = mirna_names[idx]
    cnn_important_mirnas_data.append({'mirna_name': mirna_name, 'index': idx})

# Save CNN detailed CSV
cnn_important_df = pd.DataFrame(cnn_important_mirnas_data)
cnn_important_df.to_csv('CNN_important_mirnas_list.csv', index=False)

# Save CNN simple CSV (just miRNA names)
cnn_important_simple = pd.DataFrame({'mirna': [mirna_names[idx] for idx in sorted(cnn_all_features)]})
cnn_important_simple.to_csv('CNN_important_mirnas_simple.csv', index=False)

print(f"CNN files saved: {len(cnn_important_mirnas_data)} miRNAs")
print(f"- CNN_important_mirnas_list.csv")
print(f"- CNN_important_mirnas_simple.csv")

CNN files saved: 10 miRNAs
- CNN_important_mirnas_list.csv
- CNN_important_mirnas_simple.csv

```

Task 8.1: SVM Classification on Selected miRNA Features (10/9/25)

Following Dr. Ali's instructions, I implemented Support Vector Machine (SVM) classification to evaluate the miRNA features selected from three distinct feature selection approaches: traditional methods (Chi2), deep learning (1D-CNN), and clustering (Hierarchical Clustering). For each approach, I loaded the corresponding important miRNA CSV files, extracted those specific features from the dataset, and trained separate SVM classifiers maintaining the same 70/15/15 data split protocol for consistency. The SVM models were trained with optimized hyperparameters to predict breast cancer stage classification (Stage I-IV). The classification results provided comprehensive performance metrics including accuracy, precision, recall, and F1-scores for each feature selection method. The deliverable includes documented accuracy metrics for all three approaches, confusion matrices showing classification performance across cancer stages, and trained SVM models ready for comparative analysis with other team members' classifiers.

Task 8.2: Cross-Classifier Performance Comparison (10/9/25)

Building on my SVM classification results, I will collaborate with my team members to conduct a comprehensive comparison of classifier performance across different algorithms as requested by Dr. Ali. Each team member will apply a different classifier (KNN, Random Forest, Logistic Regression, and my SVM) to the same three sets of selected miRNA features from traditional, deep learning, and clustering methods. This systematic comparison will

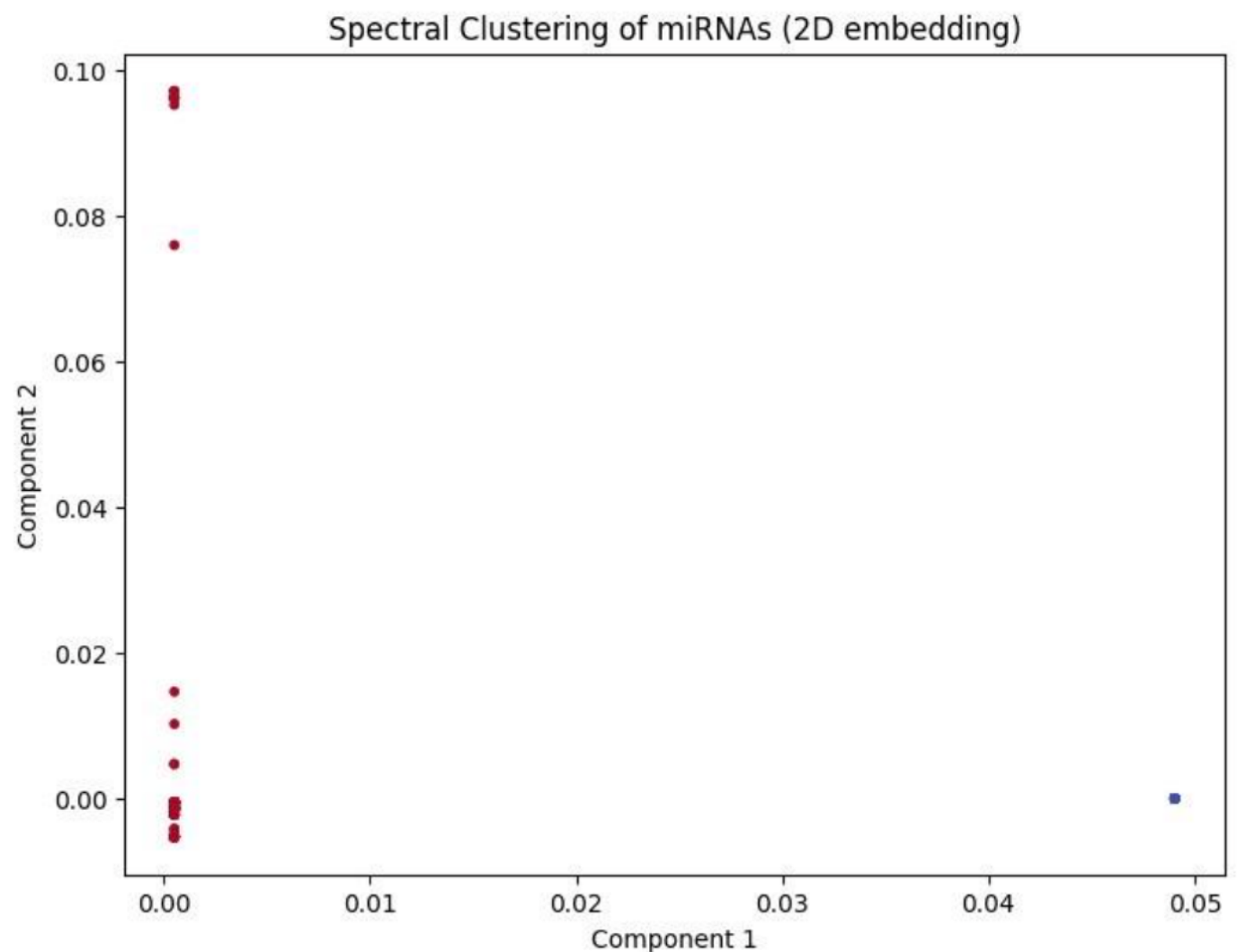
reveal which combination of feature selection method and classifier produces the best results for breast cancer stage classification. I will compile accuracy metrics from all team members' classifiers and create a comparative analysis matrix showing performance differences across methods. The analysis will identify the optimal feature selection approach by examining which method consistently produces high accuracy across multiple classifiers, and determine whether traditional statistical methods, deep learning-based selection, or unsupervised clustering yields superior features for cancer stage prediction. This comparison will directly address Dr. Ali's question about which method gives better results and provides evidence-based recommendations for the final diagnostic system implementation.

Name of the Member: **Brehon Parker**

Report Date: **9/29/25**

Task 6.1: Spectral Clustering for Unsupervised Feature Selection (9/22/25)

For this task, I applied spectral clustering to the standardized, transposed miRNA dataset in order to identify groups of miRNAs with similar expression patterns. After preparing the data by transposing and applying z-score standardization, I implemented spectral clustering with two clusters using a nearest-neighbor affinity. To evaluate the results, I generated a 2D spectral embedding plot to visualize how the features separated, and I exported the outputs into a file titled *spectral_important.csv*, which contains a ranked list of important miRNAs. This method ultimately identified roughly 800 candidate miRNAs that will serve as potential biomarkers for breast cancer stage classification. These results provide an unsupervised feature selection perspective that complements the supervised deep learning and statistical approaches from earlier weeks.



```
miRNA
hsa-mir-520e
hsa-mir-1323
hsa-mir-519b
hsa-mir-519e
hsa-mir-521-2
hsa-mir-519d
hsa-mir-1283-2
hsa-mir-498
hsa-mir-517c
hsa-mir-516b-2
hsa-mir-520b
hsa-mir-1283-1
hsa-mir-522
hsa-mir-523
hsa-mir-515-2
hsa-mir-519a-2
hsa-mir-518b
hsa-mir-520g
hsa-mir-520h
hsa-mir-520d
hsa-mir-516a-1
hsa-mir-519c
hsa-mir-520c
hsa-mir-526a-1
hsa-mir-516a-2
hsa-mir-518e
hsa-mir-518c
hsa-mir-518a-2
hsa-mir-527
hsa-mir-518a-1
hsa-mir-517b
hsa-mir-517a
hsa-mir-526b
```

Task 6.2: Preparation for Cross-Method Comparative Analysis (9/28/25)

For this task, I focused on preparing my spectral clustering results for comparison with the outputs generated by my teammates. I ensured that my spectral results were standardized in the same way to guarantee fairness in the overlap analysis. This involved confirming that the transposition and z-score normalization matched what my teammates used, validating the consistency of the input data, and checking the *spectral_important.csv* file for readiness. While the actual overlap analysis has not yet been performed, I finalized my spectral clustering results

and organized them for the next phase. The outcome of this task is that my results are now ready for the next step, which will involve a systematic overlap analysis between spectral clustering, K-means, and hierarchical clustering.

Task 7.1: Traditional (VT) and Deep Learning (DAE) Feature Outputs for Overlap Analysis (10/04/25)

Using both my Variance Threshold (traditional) and Denoising Autoencoder (deep learning) feature selection approaches, I prepared and saved the lists of significant miRNAs identified by each method. For Variance Threshold, I exported the selected features into standardized CSV outputs, formatted for consistency with the team. For DAE, I generated top-10 feature lists, ensuring they align with the simplified naming structure for later overlap comparison. These deliverables ensure that my results can be directly integrated with other team outputs (Chi2, CNN, clustering methods, etc.) to create overlap lists across traditional, deep learning, and clustering pipelines.

Task 7.2: Prepare project first demo slides (10/04/25)

For the first demo presentation, I worked on the dataset overview slide. I introduced the scope of our data — 1,209 patient samples and 1,881 miRNAs — to give context before diving into feature selection and classification. At first, I added bar and scatter plots, but after reviewing them with the team, I realized they didn't really add much value, so I removed them. Instead, I focused on keeping the slide clean, consistent with the rest of the deck, and easy to follow. My goal was to make sure the dataset was presented clearly without unnecessary clutter, so the audience can quickly understand the foundation of our work.

What has been done?



Data Preparation

- Data cleaned, standardized, and labeled with breast cancer stage information (I–IV)



Feature Selection Run

- Traditional: SFS, VT, Chi², LASSO
- Deep Learning: FNN, DAE, 1D-CNN, LSTM-Seq
- Clustering: KM, SC, GMM, HC



Overlap Lists

- Generated overlapping miRNA feature lists from traditional, deep learning, and clustering methods as input for classification



Literature Review

- Reviewed published studies and curated databases for miRNAs selected by our methods



Early Insights

- Overlaps across methods are limited, yet they highlight candidate features that may be informative for breast cancer stage classification



1,209 Samples
(Patients)



1,881 miRNAs
(Features)

Task 8.1: LR Classification on Selected miRNA Features (10/9/25)

In line with Dr. Ali's instructions, my next task will be to implement Logistic Regression (LR) classification once the three overlap lists of miRNAs are finalized (traditional, deep learning, and clustering). The plan is to extract the features from each overlap set, then train separate LR models using the established 70/15/15 data split protocol. The evaluation will focus on breast cancer stage classification (I–IV), with performance reported using accuracy, precision, recall, F1-score, and confusion matrices for each overlap set. This will allow us to directly compare how well LR performs across different feature selection pipelines, and also benchmark LR results against other classifiers chosen by the team (Random Forest, SVM, KNN).

Task 8.2: Cross-Classifer Performance Comparison (10/9/25)

For this task, once each of us has trained our assigned classifier (Logistic Regression, Random Forest, SVM, KNN) on the shared overlap lists, I'll help compare their performance side by side. The plan is to keep everything consistent by using the same 70/15/15 split and running the same set of metrics—accuracy, precision, recall, F1-score, and confusion matrices.

By lining up the results across all four classifiers and the three feature selection pipelines (traditional, deep learning, clustering), we'll be able to see which approach really stands out. This step should give us a clearer picture of the strengths and weaknesses of each method and point us toward the most promising direction for our final deliverables.

Name of the Member: Anika Baro Villa

Report Date: 9/29/25

Task 6.1: Apply K-Means clustering for feature selection (9/25/25)

I applied the K-Means clustering algorithm to group miRNAs with similar expression patterns across patients. Before clustering, I transposed and standardized the dataset to ensure each miRNA was properly scaled for distance based analysis. I ran K-Means using 2 clusters with automatic initialization, calculated cluster sizes, and evaluated the silhouette score to assess separation quality. I also visualized the results using a PCA plot to inspect the cluster distribution. I obtained an initial separation of the miRNAs into two groups, but at this stage I did not yet have enough information to determine which of these clusters was truly important.

Task 6.2: Determine the important miRNA Cluster through data-driven analysis and literature review (9/26/25)

Building on the clustering results, I carried out a data-driven analysis and a literature review to determine which of the two clusters should be treated as the “important” group for subsequent cross-method comparison. I compared the clusters using statistical measures such as variance and interquartile range on the original unscaled data and confirmed that 1 cluster consistently showed greater variability across patients (a sign that its miRNAs carry more discriminative signal). I also decided to perform a deep literature search on the miRNAs in each cluster and found that the same cluster contained a higher concentration of miRNAs previously linked to breast cancer in human studies. This combined evidence allowed me to designate that cluster (cluster 1) as the important set of features to share with the team and to serve as the basis for later comparisons.

Task 6.3: Cascade clustering applied only to the important cluster to check for finer structure (9/26/25)

After identifying the cluster of miRNAs most strongly linked to breast cancer, I applied cascade clustering to investigate whether this important group contained any further substructure. Using the same standardized data matrix but restricted to the selected miRNAs, I ran K-Means again with 2 clusters. I computed the silhouette score and visualized the results with a PCA plot to assess how clearly any subgroups emerged. The analysis revealed very low silhouette scores (around 0.02), indicating that the miRNAs within the important cluster do not form well defined subclusters under Euclidean distance. So, while the first level clustering successfully isolated a biologically relevant set of miRNAs, there is no evidence of meaningful finer partitioning within that subset.

Task 7.1: Generate a list of overlapping miRNAs across different clustering methods (9/29/25)

For this task, I compared the miRNAs within each of the “important” clusters we obtained using different clustering methods (K-Means, Spectral, Gaussian, and Hierarchical) to find overlapping features across them. K-Means’s important cluster had 464 miRNAs, Spectral’s had 46, Gaussian’s just 1 miRNA, while Hierarchical’s contained 282 miRNAs. Pairwise comparisons showed limited agreement, with overlaps only between K-Means and Hierarchical (84 miRNAs), Spectral and Hierarchical (31 miRNAs) and GMM and Hierarchical (1 miRNA). I did not find miRNAs that were common to all four methods. These results might indicate that each clustering method captures distinct data structures, with Hierarchical Clustering showing the strongest consistency with other methods. The following screenshots show part of the code, the results I obtained from this comparison, and the csv file with the overlap list of miRNAs that appear in at least 2 of the 4 clustering methods.

```
kmeans = pd.read_csv("kmeans_important.csv")['miRNA']
spectral = pd.read_csv("spectral_important.csv")['miRNA']
gmm = pd.read_csv("gaussian_cluster_1.csv")['miRNA']
hier = pd.read_csv("HC_important_mirnas_simple.csv")['miRNA']
```



```
# Comparison
print("KMeans n Spectral:", len(kmeans_set & spectral_set))
print("KMeans n GMM:", len(kmeans_set & gmm_set))
print("KMeans n Hierarchical:", len(kmeans_set & hier_set))
print("Spectral n GMM:", len(spectral_set & gmm_set))
print("Spectral n Hierarchical:", len(spectral_set & hier_set))
print("GMM n Hierarchical:", len(gmm_set & hier_set))
```

```
KMeans n Spectral: 0
KMeans n GMM: 0
KMeans n Hierarchical: 84
Spectral n GMM: 0
Spectral n Hierarchical: 31
GMM n Hierarchical: 1
```

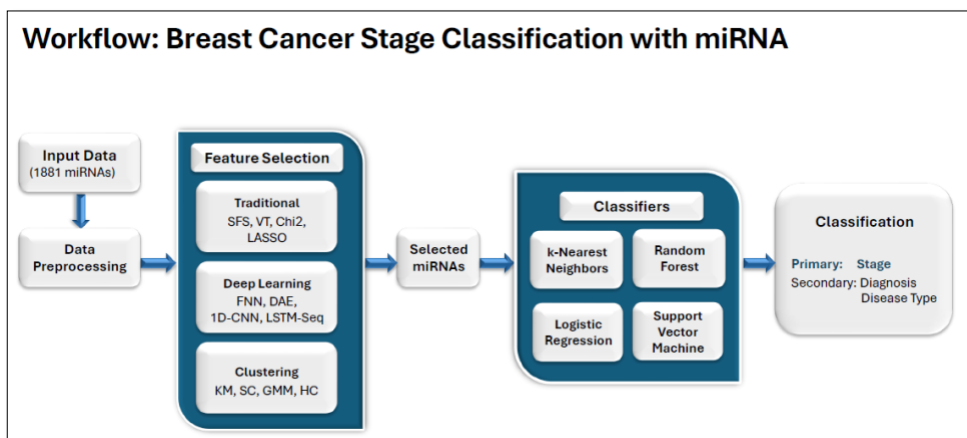
```
# Find common miRNAs to all methods
common_all = kmeans_set & spectral_set & gmm_set & hier_set
print("Number of miRNAs common to ALL methods:", len(common_all))
print("Name of miRNAs common to ALL methods:", list(common_all))
```

```
Number of miRNAs common to ALL methods: 0
Name of miRNAs common to ALL methods: []
```

consensus_clustering_overlap 9/29/2025 6:28 PM

Task 7.2: Prepare project status report slides (focus on workflow diagram) (10/03/25)

My main focus for this task was creating the workflow diagram for our breast cancer stage classification system. Drawing from our discussions with Dr. Ali Ibrahim, I designed the diagram to clearly show how all major components of the system connect and work together within the overall process. In addition to building the diagram, I revised the entire slide deck, making the necessary adjustments and additions to ensure the presentation was technically accurate and fully aligned with the current progress of our project.



Task 8.1: Increasing k for SFS and FNN to improve overlap across traditional and deep learning methods (10/12/25)

Following Dr. Ali Ibrahim's recommendation, I plan to increase the number of selected miRNAs (k) for SFS and FNN to 15 and 20 to explore whether the lack of overlap between traditional and deep learning methods is due to overly strict cutoff criteria (previously we used k=10,8,5,3). These adjustments would allow us to observe how expanding the feature sets affects the consistency and potential intersections between our methods.

Task 8.2: Apply Random Forest classification to overlapping miRNA lists (10/12/25)

I plan to apply a Random Forest classifier to the 3 overlap lists of miRNAs we obtained from the traditional, deep-learning, and clustering feature selection methods. The goal is to evaluate how well each group of shared features performs in classifying breast cancer stages and to identify which overall approach produces the most accurate and

consistent results. I will report accuracy and other performance metrics such as precision, recall, and macro-F1 score for each feature group, so these results can later be compared with the outcomes from the other classifiers used by my teammates.